

Day 41 - Syslog

Wednesday, September 9, 2026

8:37 PM

Syslog Info

Syslog General Information

Syslog is an *industry standard* protocol used for message logging on devices. It is essential when troubleshooting devices.

- **Syslog** servers listen for messages on port **514**
- It can log information such as:
 - Changes in interface status
 - Changes in OSPF neighbors
 - System restarts
 - etc.
- **Syslog** messages can be displayed in the console line, saved in the server's buffer (RAM), or sent to an external **Syslog** server.

Syslog Message Format

The Syslog message format consists of six unique values, being:

1. **Sequence (seq)**
 - A *sequence number* indicating the order/sequence of messages.
2. **Timestamp (time stamp)**
 - A *timestamp* indicating the time the message was generated.
3. **Facility (facility)**
 - A value that indicates which process on the device generated the message.
4. **Severity (severity)**
 - A number that indicates the severity of logged event.
5. **Mnemonic (mnemonic)**
 - A short code for the message, indicating what happened.
6. **Description (description)**
 - Detailed information about the event being reported.

Note on: 'seq' and 'time stamp'

The *Sequence* and *Timestamp* fields of the Syslog message may or may not be displayed, depending on the device's configuration

Syslog Severity Levels

There are 7 levels of Syslog severity, **0** being the most severe and **7** being the least severe. Here's a table of all the severity levels with a brief description:

Syslog Severity Levels Table

Level	Keyword	Description
0	Emergency	System is unusable
1	Alert	Action must be taken immediately
2	Critical	Critical conditions
3	Error	Error conditions
4	Warning	Warning conditions
5	Notification/Notice	Normal but significant condition
6	Informational	Informational messages

(0) Emergencies (1) Are (2) Critical (3) Even (4) When (5) Nobody (6) Is (7) Dying

Syslog Logging Locations

There are several different locations that Syslog messages can be sent to, including:

1. **Console line**
 - **Syslog** messages will be displayed in the CLI when connected to the device via the console port.
 - By default, all messages (level **0** - level **7**) are displayed.
2. **VTY lines**
 - **Syslog** messages will be displayed in the CLI when connected to the device via **Telnet/SSH**.
 - Requires the *terminal monitor* command to be used every time when a **Telnet/SSH** session is established in order for the messages to show
 - Disabled by default.
3. **Buffer (RAM)**
 - **Syslog** messages will be saved to the *buffer (RAM)*.
 - By default, all messages (level **0** — level **7**) are displayed.
 - You can view the messages with *show logging* command.
4. **External Server**
 - **Syslog** messages will be sent to an external server.
 - Level has be specified with the *logging trap LEVEL* command.

Syslog Configuration

- *logging console LEVEL*
 - Sets the **Syslog** logging level for *console lines*.
- *logging monitor LEVEL*
 - Sets the **Syslog** logging level for *VTY lines*.
- *logging buffered [SIZE] LEVEL*
 - Sets the **Syslog** logging level for the *buffer (RAM)*. (*size is in Bytes*)
- *logging trap TRAP-TYPES*
 - Configures the **Syslog** level for an external **Syslog** server.
- *logging IP-ADDRESS*
 - Configures a **Syslog** server that will receive **Syslog** messages.
- *logging host IP-ADDRESS*
 - Configures a **Syslog** server that will receive **Syslog** messages.
- *terminal monitor*
 - Displays Syslog messages when connected through VTY (**Telnet, SSH**).
- *logging synchronous*
 - Causes a new line to be printed if your typing is interrupted by a message. (*Must be executed in line-configuration mode*)
- *service sequence-numbers*
 - Enables *sequence numbers* to be displayed in **Syslog** messages.
- *service timestamps log TIME { DATETIME | UPTIME }*
 - Enables *timestamps* to be displayed in **Syslog** messages.
 - The '*datetime*' argument means that timestamps will display the *date/time* when the event occurred.
 - The '*uptime*' argument means that timestamps will display how long the device had been running when the event occurred.

Telnet/SSH and Syslog messages

Even if *logging monitor LEVEL* has been configured, by default, Syslog messages will not be displayed when a connection is made via Telnet/SSH.

- For messages to be displayed, you must use this command:
 - *terminal monitor* --> Displays Syslog messages when connected through **VTY (Telnet, SSH)**.
- The command must be inputted every time you connect to the device via **Telnet/SSH**

Extra: Syslog VS. SNMP

Both Syslog and SNMP are used for monitoring and troubleshooting devices. They are complementary, but their

functionalities are different.

1. Syslog

- It is used for message logging.
- Events that occur within the system are categorized based on *facility/severity* and logged.
- Used for system management, analysis, and troubleshooting.
- Messages are sent from the devices to the server. The server can't actively pull information from the devices (like **SNMP Get**) or modify variables (like **SNMP Set**).

2. SNMP

- Used to retrieve and organize information about the **SNMP managed devices**, such as:
 - IP addresses, current interface status, temperature, CPU usage, etc.
- **SNMP** servers can use *Get* to query the clients and *Set* to modify variables on the clients.